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ABSTRACT

This report uses important mathematical techniques such as Linear Algebra, Calculus, Probability and Statistics to analyse elements of a smart urban mobility system in Manchester. The aim is to show how mathematical modelling can help data driven decision making in transport systems.

Traffic flow across a network of junctions is modelled using Linear algebra and a system of simultaneous equations. Solutions of these equations are used to find the equilibrium conditions in the network. Calculus is used to derive acceleration and to determine changes in vehicle speed with time. The work provides some insights into the traffic efficiency and driver behaviour. Then the uncertainty of autonomous vehicle response time is modelled using probability and a normal distribution to estimate the probability of fast and slow reaction times. Then statistical methods are applied to observe vehicle speed data in an Intelligent Speed Assistance (ISA) zone, where the compliance with the legal speed limit is assessed using descriptive statistics and hypothesis testing.

The obtained results show that there is a stable traffic flow but some roads are more in demand so they are more likely to get congested. The acceleration analysis shows fluctuations which indicate inefficiencies in the traffic flow. The probability analysis shows the importance of taking into account the variability of reaction times for safety critical systems. Statistical results shows that the average vehicle speed is higher than the legal limit, which indicates that the ISA system may have limitations.

The report demonstrates the importance of mathematical methods in providing useful insights into traffic behaviour, safety and system performance. However the analysis relies on simplified assumptions and limited data, and future work should include more realistic data samples and advanced modelling techniques to improve accuracy and implementation in smart mobility systems.

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URBAN MOBILITY INTELLIGENCE UNIT: REPORT OF SMART MOBILITY ANALYSIS IN MANCHESTER

1. INTRODUCTION

This report includes investigative results after the analysis of different aspects of urban traffic behaviour and system performance in Manchester and provides insights into how EV charging infrastructure affects equilibrium traffic flows, how vehicle speeds and acceleration patterns vary along smart corridors, how uncertainty in autonomous vehicle reaction times can be quantified, and whether observed vehicle speeds comply with regulatory limits in ISA-equipped zones.

Smart mobility systems are gaining importance in urban transport planning. For instance, in Manchester since the implementation of adaptive traffic signals, electric vehicle charging corridors, autonomous vehicle technologies and intelligent speed assistant systems the road transport networks became more complex. Even though they are implemented to reduce congestion, improve safety and make the traffic flow efficient they also create a need to properly analyse traffic behaviour. (Ali *et al.*, 2026)

This report shows the results driven after using mathematical and computational methods on traffic data to analyse the traffic behaviour so as to provide a basis for understanding transport performance and as a result to support data driven decision making.

1.1. Data Description

The data available represents different aspects of smart mobility system in Manchester and are as follows:

- A system of equations which describes traffic flow across multiple junctions.
- A time dependant velocity function which describes vehicle movement along a smart corridor.
- A probability model which describes autonomous vehicle reaction times.
- A sample dataset of observed vehicle speeds within a regulated zone.

1.2. Analytical Approach

Different mathematical approaches that will be used to analyse the above data are as follows:

- Linear algebra - to solve the system of equations to see if a stable traffic flow exists.
- Calculus - to study how the speed of vehicle fluctuates at different times and to recognise braking and acceleration patterns.
- Probability - to analyse uncertainty in the behaviour of autonomous vehicles.
- Statistics - to check whether the observed speed of vehicles falls within the legal speed limit.

By the combined use of the above methods, the available data can be understood better and be interpreted in a way that allows us to make meaningful conclusions about the performance of the traffic system and safety.

2. DATA ANALYSIS

In the following subsections the analytical approach mentioned above are implemented with clear explanations and results are obtained after carrying out each of the mathematical methods on the data.

2.1. Linear Algebra: Identifying if traffic flow equilibrium exists or not

Traffic networks can be represented using a system of simultaneous linear equations, hence Linear Algebra is the most appropriate method to model traffic flow.

2.1.1. System Description

The traffic system under our investigation is represented by four unknown variables:

$$x_1, x_2, x_3, x_4$$

Each unknown variable represents traffic flow along four key roads, measured in vehicles per 10 minutes.

The system of equations are as follows:

$$2x_1 - x_2 + x_3 = 150$$

$$x_1 + x_2 + x_4 = 420$$

$$-x_1 + 3x_3 - x_4 = 100$$

$$x_2 + x_3 + x_4 = 380$$

Linear algebra is now used to determine whether the network of interconnected junctions has a steady state equilibrium.

These equations are written in matrix form as (Strang, 2016):

$$AX = B$$

Where,

$$A = \begin{bmatrix} 2 & -1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ -1 & 0 & 3 & -1 \\ 0 & 1 & 1 & 1 \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad B = \begin{bmatrix} 150 \\ 420 \\ 100 \\ 380 \end{bmatrix}$$

2.1.2. Solution

Solution using Gaussian Elimination:

Row operations are applied to change the augmented matrix into row echelon form and then we can use back substitution to determine the unknown variables (Lay, 2012).

Augmented matrix:

$$\left[\begin{array}{cccc|c} 2 & -1 & 1 & 0 & 150 \\ 1 & 1 & 0 & 1 & 420 \\ -1 & 0 & 3 & -1 & 100 \\ 0 & 1 & 1 & 1 & 380 \end{array} \right]$$

Swap rows to make 1 leading:

$$R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 420 \\ 2 & -1 & 1 & 0 & 150 \\ -1 & 0 & 3 & -1 & 100 \\ 0 & 1 & 1 & 1 & 380 \end{array} \right]$$

Remove x_1 from other rows:

$$R_2 \leftarrow R_2 - 2R_1 \quad R_3 \leftarrow R_3 + R_1$$

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 420 \\ 0 & -3 & 1 & -2 & -690 \\ 0 & 1 & 3 & 0 & 520 \\ 0 & 1 & 1 & 1 & 380 \end{array} \right]$$

Swapping R_2 and R_3 :

$$\left[\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 420 \\ 0 & 1 & 3 & 0 & 520 \\ 0 & -3 & 1 & -2 & -690 \\ 0 & 1 & 1 & 1 & 380 \end{array} \right]$$

Remove x_2 :

$$R_1 \leftarrow R_1 - R_2 \quad R_3 \leftarrow R_3 + 3R_2 \quad R_4 \leftarrow R_4 - R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 1 & -100 \\ 0 & 1 & 3 & 0 & 520 \\ 0 & 0 & 10 & -2 & 870 \\ 0 & 0 & -2 & 1 & -140 \end{array} \right]$$

Make pivot in R_3 as 1:

$$R_3 \leftarrow \frac{1}{10} R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 1 & -100 \\ 0 & 1 & 3 & 0 & 520 \\ 0 & 0 & 1 & -0.2 & 87 \\ 0 & 0 & -2 & 1 & -140 \end{array} \right]$$

Remove x_3 :

$$R_4 \leftarrow R_4 + 2R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 1 & -100 \\ 0 & 1 & 3 & 0 & 520 \\ 0 & 0 & 1 & -0.2 & 87 \\ 0 & 0 & 0 & 0.6 & 34 \end{array} \right]$$

Now using back substitution:

$$\text{From row 4: } 0.6x_4 = 34 \Rightarrow x_4 = 56.67$$

$$\text{From row 3: } x_3 - 0.2x_4 = 87 \Rightarrow x_3 = 98.33$$

$$\text{From row 2: } x_2 + 3x_3 = 520 \Rightarrow x_2 = 225$$

$$\text{From row 1: } x_1 - 3x_3 + x_4 = -100 \Rightarrow x_1 = 138.33$$

Hence,

$$x_1 = 138.33, x_2 = 225, x_3 = 98.33, x_4 = 56.67$$

2.1.3. Interpretation of Results

The solutions shows that a stable equilibrium of traffic flow exists along the four roads. All the values of the unknown variables are found to be positive, so the solution is meaningful and shows us that a stable steady state flow exists within the system. Also the determinant of Matrix A is not zero, showing us that the system has a unique solution.

Additionally we can see that the traffic is the highest on road x_2 and this is the road that will be affected more by congestion due to EV charging or merging behaviour. Meanwhile traffic is the least on road x_4 .

2.2. Calculus: Evaluating vehicle speed behaviour

Calculus is used in Data Science to see how quantities change over time. (Arden University, 2025) In context of the system under consideration, it helps to understand how vehicle speed varies and to understand the acceleration and breaking patterns.

2.2.1. System Description

The velocity of vehicles travelling along the smart mobility corridor is given by the function:

$$v(t) = 70 + 12 \cos\left(\frac{\pi t}{5}\right) - 4 \sin\left(\frac{2\pi t}{5}\right)$$

where:

t is time in seconds

$v(t)$ is velocity in km/h

2.2.2. Solution

To see how speed changes the acceleration function should be determined, this is done using differentiation. Acceleration is rate of change of velocity with time. (Stewart, 2016)

$$a(t) = \frac{dv}{dt}$$

Differentiate constant:

$$\frac{d}{dt}(70) = 0$$

Differentiate cosine term (using chain rule):

$$\begin{aligned}\frac{d}{dt} \left[12 \cos \left(\frac{\pi t}{5} \right) \right] &= 12 \cdot \left(-\sin \left(\frac{\pi t}{5} \right) \right) \cdot \frac{\pi}{5} \\ &= -\frac{12\pi}{5} \sin \left(\frac{\pi t}{5} \right)\end{aligned}$$

Differentiate sine terms:

$$\begin{aligned}\frac{d}{dt} \left[-4 \sin \left(\frac{2\pi t}{5} \right) \right] &= -4 \cdot \cos \left(\frac{2\pi t}{5} \right) \cdot \frac{2\pi}{5} \\ &= -\frac{8\pi}{5} \cos \left(\frac{2\pi t}{5} \right)\end{aligned}$$

Final acceleration function:

$$a(t) = -\frac{12\pi}{5} \sin \left(\frac{\pi t}{5} \right) - \frac{8\pi}{5} \cos \left(\frac{2\pi t}{5} \right)$$

This describes how the vehicle speed changes over time. Since the function includes both sine and cosine terms it means the function is periodic, that is the behaviour is repeated over time.

Numerical evaluation was done in Python to identify the maximum and minimum acceleration values. For that a range of time values in between 0 and 10 seconds was generated, and the acceleration function was evaluated at each point

Obtained results:

Maximum acceleration ≈ 12.57 happening at around 7.5 seconds

Minimum acceleration ≈ -6.44 happening at around 0.61 seconds

2.2.3. Interpretation of Results

The obtained values shows that the movement of vehicles along the corridor is not constant and it changes over time.

The large positive values obtained indicates rapid acceleration and this might lead to

- Discomfort for passengers
- More energy consumption

The large negative values obtained indicates strong braking and this possibly shows

- Traffic jam
- Improper signal timing

If solutions for the above mentioned problems can be implemented, it will help in

- Reducing traffic jams and improves the flow
- Increasing the passenger comfort
- Reducing fuel consumptions and emissions

2.3 Probability: Analysing uncertainty in the behaviour of autonomous vehicles

Probability allows uncertainty in real world systems to be quantified and analysed. In the Manchester smart mobility system under consideration, the probability of uncertain outcomes plays a very important role while considering autonomous vehicle reaction times, since delays may affect road safety.

2.3.1. System Description

The reaction time R of an autonomous vehicle is modelled as a normally distributed random variable:

$$R \sim N(\mu = 0.85 \text{ s}, \sigma = 0.25 \text{ s})$$

$\mu = 0.85$ seconds (mean reaction time)

$\sigma = 0.25$ seconds (standard deviation)

2.3.2. Solution

Standardisation using Z-Score:

For a normal distribution, to find the probability, the data has to be standardised using a z-score. (Montgomery and Runger, 2014)

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Defining two thresholds for reaction times:

Rapid reaction: $R < 0.60$ seconds

Delayed reaction: $R > 1.10$ seconds

Probability of Rapid Reaction:

$$Z = \frac{0.60 - 0.85}{0.25}$$

$$Z = \frac{-0.25}{0.25}$$

$$Z = -1$$

Using standard normal distribution:

$$P(R < 0.60) = P(Z < -1)$$

$$P(Z < -1) = 0.1587$$

Probability of Delayed Reaction:

$$Z = \frac{1.10 - 0.85}{0.25} = 1$$

Using standard normal distribution:

$$P(R > 1.10) = P(Z > 1)$$

$$P(Z < 1) = 0.8413$$

$$P(Z > 1) = 1 - P(Z < 1)$$

$$P(Z > 1) = 1 - P(Z < 1)$$

$$P(Z > 1) = 1 - 0.8413$$

$$P(Z > 1) = 0.1587$$

Hence:

Probability of rapid reaction = 15.87%

Probability of delayed reaction = 15.87%

Percentile Calculation:

Formula:

$$X = \mu + Z\sigma$$

5th percentile:

$$P(R < X) = 0.05$$

$$Z = -1.645$$

$$X = 0.85 + (-1.645)(0.25)$$

$$X = 0.85 - 0.41125$$

$$X = 0.43875$$

$$X \approx 0.44$$

So approximately 5% of reaction times are found to be below 0.44 seconds

95th percentile:

$$P(R < X) = 0.95$$

$$Z = 1.645$$

$$X = 0.85 + (1.645)(0.25)$$

$$X = 0.85 + 0.41125$$

$$X = 1.26125$$

$$X \approx 1.26$$

So approximately 95% of reaction times are found to be below 1.26 seconds

2.3.3. Interpretation of Results

The obtained results suggests that most vehicle reaction times are expected to be close to the mean which is 0.85 seconds. If the reaction time is less than 0.60 seconds it is comparatively fast and if the reaction time is more than 1.10 seconds it is comparatively delayed. Both are found to have the probability of 15.87%.

The most expected reaction time is 1.26 seconds as we found out by defining the 95th percentile.

2.4. Statistics: ISA speed compliance analysis

Statistics in Data Science is used to summarise data and evaluate evidence and make decisions based on it. In this section statistics is used to check whether the observed vehicle speeds falls under the legal speed limit.

2.4.1. System Description

The observed vehicle speeds are:

49, 52, 55, 48, 50, 53, 57, 51, 54, 49, 52, 56

Legal speed limit:

50 km/h

2.4.2. Solution

Descriptive Statistics (Arden University, 2025)

Number of observations:

$$n = 12$$

Mean:

$$\bar{x} = \frac{\sum x}{n}$$

$$\bar{x} = \frac{626}{12}$$

$$\bar{x} = 52.17 \text{ km/h}$$

Median:

$$\text{Median} = \frac{52+52}{2}$$

$$\text{Median} = 52 \text{ km/h}$$

Sample standard deviation:

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$$s \approx 2.92 \text{ km/h}$$

Sample variance:

$$s^2 = 2.92^2$$

$$s^2 \approx 8.52 \text{ km/h}$$

Hypothesis testing (Arden University, 2025)

The null hypothesis is:

$$H_0 = \mu = 50 \text{ (true mean vehicle speed is equal to legal speed limit)}$$

Alternative hypothesis:

$$H_1 = \mu > 50 \text{ (true mean vehicle speed is greater than legal speed limit)}$$

T-test Formula: (Montgomery and Runger, 2014)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
$$t = \frac{52.17 - 50}{2.92/\sqrt{12}}$$
$$t = \frac{2.17}{0.843}$$
$$t = 2.57$$

Degrees of freedom:

$$df = n - 1$$
$$df = 11$$

P value:

$$p = 0.013 \text{ (using python)}$$
$$\text{since } p < 0.05$$

we reject null hypothesis (H_0)

This shows that mean vehicle speed is greater than legal limit of 50km/h

2.4.3. Interpretation of Results

The results shows that the vehicles in the ISA – equipped zone are not fully complying with the speed limit. This might be because the speed control by the Intelligent Speed Assistance system is not proper or because the drivers are careless.

3. CONCLUSION

3.1. Results and Discussion

The mathematical analysis provides important insights into different aspects of Smart Urban Mobility system in Manchester. Each mathematical method gives a different view to understanding the system behaviour.

The linear algebra analysis shows that the traffic network has a unique equilibrium solution with traffic flows of approximately 138.33, 225.00, 98.33 and 56.67 vehicles in every 10 minutes. This shows that it is a stable system under the assumptions made. However the results shows that there is high traffic volume on some route, like route x_2 . So this route might be more congested than the other routes and might be more affected by EV charging, merging or stopping behaviour.

The calculus analysis gave an idea of how the vehicle speed changes over time. The acceleration profile was obtained by differentiating the velocity function and it showed both positive and negative fluctuations. The maximum acceleration of about 12.57 and minimum acceleration of about -6.44 indicate that the vehicles undergo a period of rapid change of speed. This indicates that the corridor may show stop-start traffic behaviour, possibly due to traffic signals or traffic jams. Such variations can cause passenger discomfort and increased energy consumption. So these shows some inefficiencies in the traffic system.

The probability analysis showed that the reaction times of the vehicles are normally distributed with a mean of 0.85 seconds. It is also found that the probability of rapid and delayed reactions is about 15.87%. Also from the percentile analysis it is clear that most reaction times ranged from 0.44 to 1.26 seconds. This shows how important it is to take into account the variability of reaction times in the design of such systems rather than just the average. (Montgomery and Runger, 2014; Saini, 2025)

The statistics analysis showed that the average observed vehicle speed in the ISA zone is 52.17 km/h which is greater than the legal speed limit of 50 km/h. The hypothesis test showed that this difference was statistically significant. These results suggests that the Intelligent Speed Assistance system might not guarantee complete compliance and it is better to implement additional measures to improve speed control.

Overall, the obtained results shows that even though smart mobility systems provide valuable data and control mechanisms there are still some challenges regarding congestion, efficiency, safety and compliance. The Mathematical methods used helps in identifying these challenges and in decision making.

3.2. Reflective Report

The mathematical methods used above to study the different components of a smart mobility system are efficient however each method is based on certain assumptions and has some limitations that we have to take into consideration.

The linear algebra model is based on the assumption that the traffic flow is constant and the system is in steady state. But in reality traffic flows changes over time as a result of rush hour, accidents, weather etc. So, even if the model is useful to understand the equilibrium behaviour, it may not be able to fully understand the traffic patterns. So a possible improvement that can be made is to add time dependent models or simulation based models to better represent the real world traffic.

In the calculus model, we assume that the velocity function accurately represents the vehicle movement. This is also a simplified function and may not include real world factors like drivers behaviour, road conditions, unexpected disruptions etc. Differentiation does provide some information about the nature of the acceleration, but does not consider the randomness of the system. Future work in this may involve real sensor data or more data driven models for more accuracy.

The probability model assumes that the reaction times are normally distributed, but real data may not perfectly follow this distribution. For example, rare but critical delayed reactions may not be completely captured. To improve this other distributions could be tested or the model could be validated using empirical data.

The statistical analysis uses a very small sample size of 12 observations. The t-test performed is suitable for small samples but may not give proper representation of the speed of vehicles for an entire population. Larger data set would make the results more reliable. Here also the analysis does not consider external factors such as driver behaviour, road conditions etc which may affect the result as well.

However the mathematical methods used in this study has many other positive sides as well. They provide clear structured methods for analysing complex systems and allow meaningful conclusions to be drawn from small amounts of data. The combination of Linear Algebra, Calculus, Probability and Statistics gives a proper understanding of different aspects of smart mobility systems.

4. REFERENCES

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5. APPENDIX

Linear Algebra:

```
[ ] import numpy as np
import matplotlib.pyplot as plt

[ ] A = np.array([
    [2, -1, 1, 0],
    [1, 1, 0, 1],
    [-1, 0, 3, -1],
    [0, 1, 1, 1]
], dtype=float)

[ ] B = np.array([150, 420, 100, 380], dtype=float)

[ ] x = np.linalg.solve(A, B)

print("Traffic flow solution:")
print("x1 =", x[0])
print("x2 =", x[1])
print("x3 =", x[2])
print("x4 =", x[3])

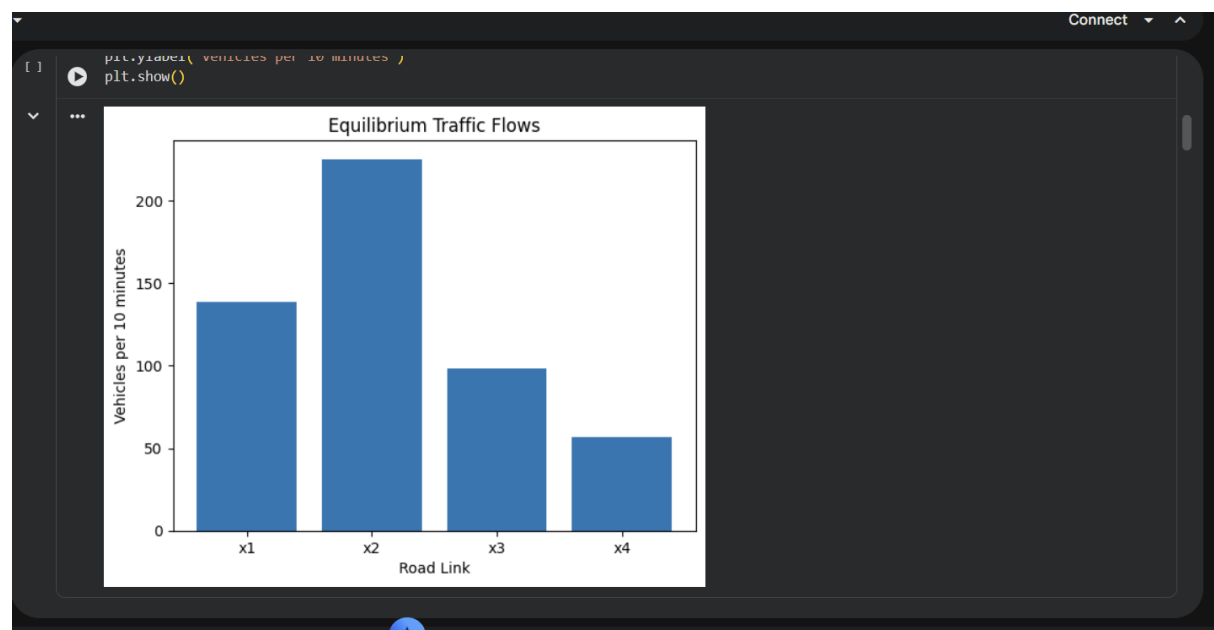
Traffic flow solution:
x1 = 138.33333333333334
x2 = 225.0
x3 = 98.33333333333333
x4 = 56.66666666666668
```

```
[ ] det_A = np.linalg.det(A)
print("Determinant of A =", det_A)

Determinant of A = 6.000000000000003

[ ] labels = ["x1", "x2", "x3", "x4"]

[ ] plt.bar(labels, x)
plt.title("Equilibrium Traffic Flows")
plt.xlabel("Road Link")
plt.ylabel("Vehicles per 10 minutes")
plt.show()
```



Calculus:

```
CALCULUS

[ ] import numpy as np

[ ] t = np.linspace(0, 10, 1000)

[ ] v = 70 + 12*np.cos(np.pi*t/5) - 4*np.sin(2*np.pi*t/5)

[ ] a = -(12*np.pi/5)*np.sin(np.pi*t/5) - (8*np.pi/5)*np.cos(2*np.pi*t/5)

[ ] max_acc = np.max(a)
    min_acc = np.min(a)

[ ] t_max = t[np.argmax(a)]
    t_min = t[np.argmin(a)]

[ ] print("Maximum acceleration:", max_acc)
    print("Time at max acceleration:", t_max)

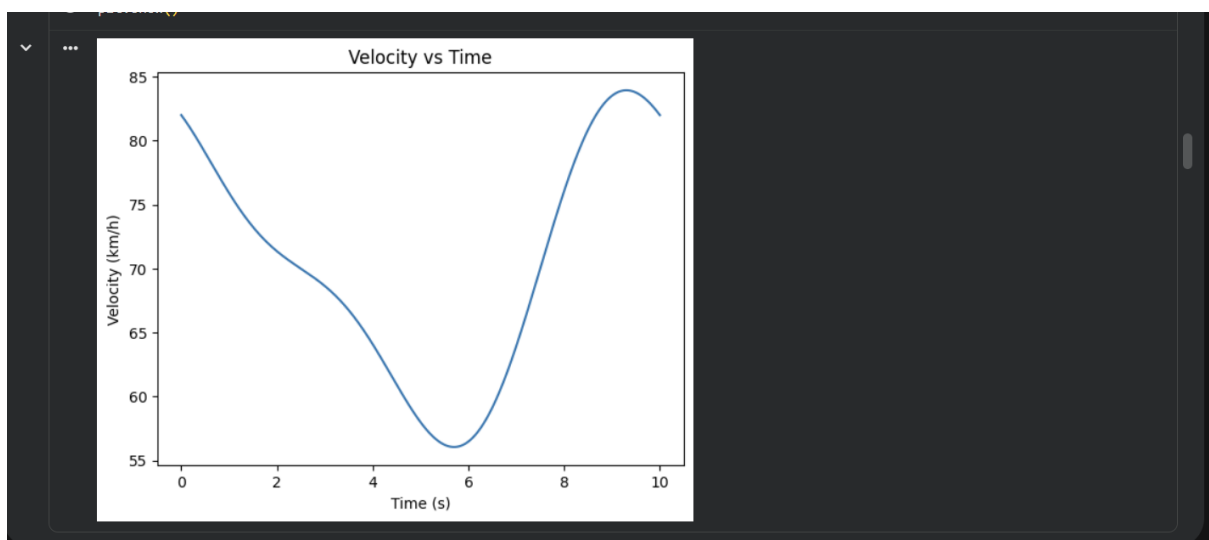
... Maximum acceleration: 12.566336439160962
    Time at max acceleration: 7.4974974974974975

[ ]
```

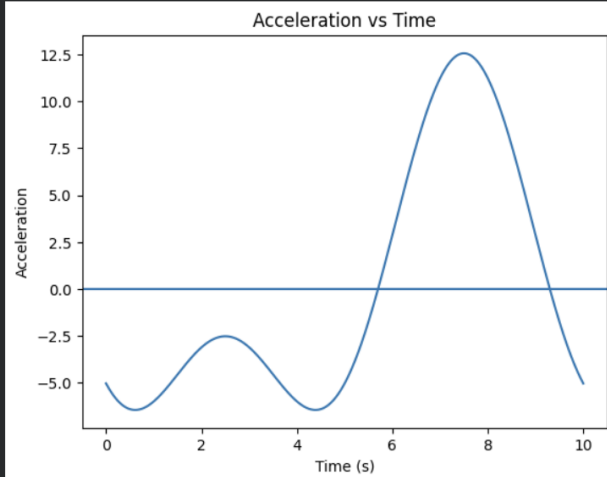
```
[ ] print("Minimum acceleration:", min_acc)
    print("Time at min acceleration:", t_min)

... Minimum acceleration: -6.440260222693828
    Time at min acceleration: 0.6106106106106106

[ ] plt.plot(t, v)
    plt.title("Velocity vs Time")
    plt.xlabel("Time (s)")
    plt.ylabel("Velocity (km/h)")
    plt.show()
```



```
[ ] ▶ plt.plot(t, a)
plt.title("Acceleration vs Time")
plt.xlabel("Time (s)")
plt.ylabel("Acceleration")
plt.axhline(0)
plt.show()
```



Probability:

PROBABILITY

```
[ ] import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

[ ] mu = 0.85
sigma = 0.25

[ ] p_rapid = stats.norm.cdf(0.60, loc=mu, scale=sigma)

[ ] p_delayed = 1 - stats.norm.cdf(1.10, loc=mu, scale=sigma)

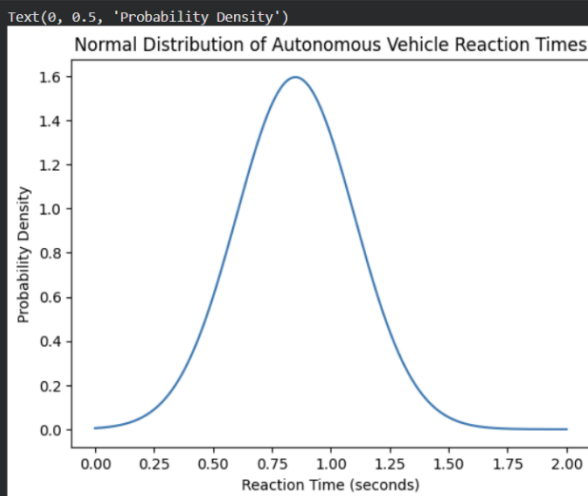
[ ] percentile_5 = stats.norm.ppf(0.05, loc=mu, scale=sigma)
percentile_95 = stats.norm.ppf(0.95, loc=mu, scale=sigma)

[ ] print("Probability of rapid reaction P(R < 0.60):", p_rapid)
print("Probability of delayed reaction P(R > 1.10):", p_delayed)
print("5th percentile:", percentile_5)
print("95th percentile:", percentile_95)
```

```
Probability of rapid reaction  $P(R < 0.60)$ : 0.15865525393145707  
Probability of delayed reaction  $P(R > 1.10)$ : 0.15865525393145696  
5th percentile: 0.43878659326213176  
95th percentile: 1.261213406737868
```

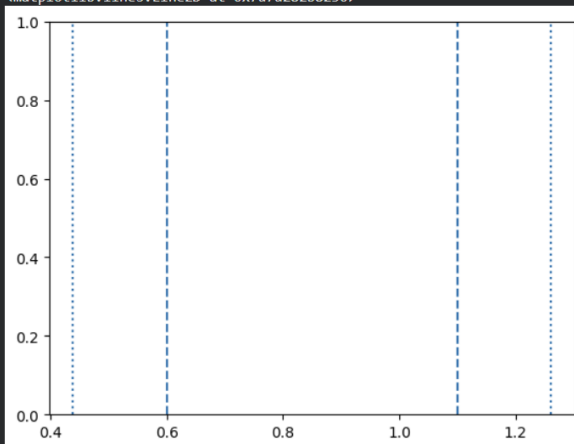
```
[ ] x = np.linspace(0, 2, 1000)  
y = stats.norm.pdf(x, loc=mu, scale=sigma)
```

```
[ ] ▶ plt.plot(x, y)  
plt.title("Normal Distribution of Autonomous Vehicle Reaction Times")  
plt.xlabel("Reaction Time (seconds)")  
plt.ylabel("Probability Density")
```



```
[ ] ▶ plt.axvline(0.60, linestyle="--", label="Rapid threshold: 0.60s")  
plt.axvline(1.10, linestyle="--", label="Delayed threshold: 1.10s")  
plt.axvline(percentile_5, linestyle=":", label="5th percentile")  
plt.axvline(percentile_95, linestyle=":", label="95th percentile")
```

```
▼ ... <matplotlib.lines.Line2D at 0x7d7d282b8230>
```



Statistics:

```
STATISTICS

[ ] import numpy as np
    from scipy import stats
    import matplotlib.pyplot as plt

[ ] speeds = np.array([49, 52, 55, 48, 50, 53, 57, 51, 54, 49, 52, 56])

[ ] n = len(speeds)
    mean_speed = np.mean(speeds)
    median_speed = np.median(speeds)
    std_speed = np.std(speeds, ddof=1)
    var_speed = np.var(speeds, ddof=1)

[ ] print("Sample size:", n)
    print("Mean speed:", mean_speed)
    print("Median speed:", median_speed)
    print("Sample standard deviation:", std_speed)
    print("Sample variance:", var_speed)

▼ Sample size: 12
   Mean speed: 52.166666666666664
   Median speed: 52.0
   Sample standard deviation: 2.9180732539042804
   Sample variance: 8.515151515151514
```

```
[ ] legal_limit = 50

[ ] t_stat, p_two_tailed = stats.ttest_1samp(speeds, popmean=legal_limit)

[ ] p_one_tailed = p_two_tailed / 2

[ ] print("t-statistic:", t_stat)
    print("One-tailed p-value:", p_one_tailed)

▼ t-statistic: 2.572092215102194
   One-tailed p-value: 0.012976731760515154

[ ] plt.hist(speeds, bins=6)
    plt.axvline(50, linestyle="--", label="Legal limit: 50 km/h")
    plt.axvline(mean_speed, linestyle=":", label="Mean speed")
    plt.title("Observed Vehicle Speeds in ISA Zone")
    plt.xlabel("Speed (km/h)")
    plt.ylabel("Frequency")
    plt.legend()
    plt.show()
```

